

Does Cedar Loop's pickup delay rate warrant immediate permanent staffing or a bounded audit pilot?

Your guide for the lab. Start with the email, then work each card in order. Everything else is optional reference.

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The request

Subject: **Does Cedar Loop's pickup delay rate warrant immediate permanent staffing or a bounded audit pilot?**



VP of Retail Operations

To: Lead Operations Analyst (you)

Hi Lead Operations Analyst,

Whether to authorize permanent evening fulfillment staff based on audit sample delays or mandate a bounded verification pilot before spending capital.

Specifically, I need answers to 4 questions:

1. Sampling Frame and Nonresponse Audit

Part A

2. Duration Estimation and Confidence Intervals

Part B

3. Proportion Confidence Interval and Sample Size Planning

Part C

4. Post-Class Reflection and Decision Revisit

Take-home

For this practice:

- Keep your completed workbook for discussion and your own review.
- This is ungraded practice. No memo, upload, submission or deadline is required.

Thank you.

Best,

VP of Retail Operations

Cedar Loop Grocery Pickup



cedar_loop_pickups.csv



S05P_Learning_Workbook.xlsx

Your lab at a glance

Before class

setup
untimed

Part A

10 min
Slide 28 → 29

Part B

10 min
Slide 46 → 47

Part C

10 min
Slide 66 → 70

Take-home

15 min
At home

Finish

checklist
checklist

Before class: get ready (untimed)

- 1** Unzip `S05P_Student_Package.zip` into one folder.
Keep every file together. SPSS looks for the CSV files next to the syntax.
- 2** Open `S05P_Learning_Workbook.xlsx` and go to the **Start Here** sheet.
Type your first name, last name and NetID (the part before @nyu.edu). Use **File > Save As** with the workbook name it shows.
- 3** Open `S05P_SPSS.sps` in SPSS and update the `CD` command.
Replace `REPLACE_WITH_FOLDER_CONTAINING_CSV_AND_SPS` with the path of your unzipped lab folder (the folder, not the `.sps` file; see [Find your folder path](#)). Save As the `.sps` name shown on Start Here.
- 4** Select the whole **H0** block, then choose **Run > Selection**.
The Output Viewer opens with **H0 Import and Frame Check**.

✅ **Ready when:** the H0 output shows the expected case count and your workbook is saved under your own name.

⚠️ **Stuck?** Don't start Part A on the timer. Ask for help first, or see the SPSS environment guide.

Find your folder path (for the `CD` line)

SPSS needs the path of the **folder** you unzipped, the one that holds `S05P_SPSS.sps` and the CSV files. It is **not** the path to the `.sps` file. If you open the folder and see the `.sps` file and the CSV files inside, that is the right folder.

Windows

1. Unzip first: right-click `S05P_Student_Package.zip` > **Extract All** > **Extract**. Do not work inside the zip.
2. Open the extracted `S05P_Student_Package` folder, then the **Student** folder inside it. You should now see `S05P_SPSS.sps` and the CSV files. This is your lab folder.
3. Click once in the address bar at the top of File Explorer: it turns into the path. Press **Ctrl+C**.
4. Or: go up one level, hold **Shift**, right-click that folder > **Copy as path**, then delete the double quotes it adds.

Looks like

```
C:\Users\you\Downloads\S05P_Student_Package\Student
```

Mac

1. Unzip first: double-click `S05P_Student_Package.zip` in Finder.
2. Open the extracted `S05P_Student_Package` folder, then the **Student** folder inside it. You should now see `S05P_SPSS.sps` and the CSV files. This is your lab folder.
3. Go up one level and click that folder **once** to select it (the folder, not a file inside it). Press **Option+Command+C** (Copy as Pathname), or right-click it, hold **Option** and choose **Copy ... as Pathname**.

Looks like

```
/Users/you/Downloads/S05P_Student_Package/Student
```

Paste it between the single quotes on the `CD` line:

```
CD 'C:\Users\you\Downloads\S05P_Student_Package\Student'.  
CD '/Users/you/Downloads/S05P_Student_Package/Student'.
```

* Windows

* Mac

⚠ **Common mistakes:** the path ends in `/S05P_SPSS.sps` (remove the file name); the quotes or the final period are missing; the files are still inside the zip; OneDrive or iCloud has not finished downloading the folder. If H0 says a file cannot be found, check the path first. If the path itself contains an apostrophe (`'`), put the path in double quotes instead, for example `CD "/Users/you/Kim's files/S05P_Student_Package/Student"`.

How every card works

Yellow cells: you type here Lavender cells: guidance

✓ **Check your work:** in-class boxes turn to "✓ Checked"; take-home boxes read "Entry recorded" once each answer is a formula

Each card opens with a short recap of the session content it uses, with the slides and Toolkit pages to reread. Type each formula starting with `=`, then press **Enter**. The cells are already formatted, so never multiply a percentage by 100.

Part A · Sampling Frame and Nonresponse Audit

From slide 28 → work this card → Back to slide 29

· bring: The nonresponse sensitivity range [19.2%, 39.2%] and its relation to the 20% target.

WHAT THE SESSION COVERED · SECTION 1 SAMPLING DESIGN, BIAS, AND THE LIMITS OF GENERALIZATION

- **Target Population and Sampling Frame:** A target population is the complete group of decision interest, while the sampling frame is the concrete operational list used to draw units. *Assuming an internal transactional database automatically includes all customers without duplicate or canceled orders.*
- **Nonresponse Sensitivity Bounds:** Nonresponse sensitivity bounds establish extreme best-case and worst-case scenario intervals for unobserved units in an audited sample. *Treating the 24.0% respondent rate as the true rate while ignoring the 25 missing outcomes.*

COURSE SLIDES

Slides 9–28

Target Population and Sampling Frame 10 · Sample Proportion Point Estimate 12–14 · Excel Draw: Frozen Keys 15–17 · Probability Sampling 18 · Selection bias, nonresponse, and independence... 20 · Nonresponse Sensitivity Scenario Bounds 21–23

CONCISE TOOLKIT

pp. 3–9

Target Population and Sampling Frame p. 3 · Nonresponse Sensitivity Bounds p. 6 · Cedar Loop Grocery Pickup p. 6

FULL TOOLKIT

pp. 4–24

Target Population and Sampling Frame p. 21 · Nonresponse Sensitivity Bounds p. 16 · Cedar Loop Grocery Pickup p. 16

THE RULES YOU ARE APPLYING

TARGET POPULATION AND SAMPLING FRAME

N = Total audited frame units

NONRESPONSE SENSITIVITY BOUNDS

$$p_{\min} = \frac{x_{\text{resp}}}{n_{\text{selected}}}, \quad p_{\max} = \frac{x_{\text{resp}} + m}{n_{\text{selected}}}$$

FIRST, THE SETUP · ONCE PER SPSS SESSION

Run this **once each time you open SPSS**, before any part. On the **CD** line, replace

REPLACE_WITH_FOLDER_CONTAINING_CSV_AND_SPS with the path of your **unzipped lab folder** (keep the single quotes and the final period). It is the folder that holds **S05P_SPSS.sps** and the data files, **not** the **.sps** file itself: see **Find your folder path**. Select from the first line down to the end of the H0 block and choose **Run > Selection**. Already ran it since you opened SPSS? Skip to the block for this part.

* *Lecture connection: Target Population and Sampling Frame .*

* *Why: Validate and import the one provisional pickup roster before any dependent analysis .*

* *How: CD points to the folder containing the supplied CSV; GET DATA preserves the ID as text and imports numeric blanks as missing .*

* *How: The supplied validation rejects malformed nonblank numeric tokens and unexpected binary values before import .*

* *Read the result: H0 invitation frequencies account for the 2,400 recorded rows; that count does not certify frame completeness .*

** Watch out: Running H1-H3 before H0 or pointing CD at the syntax file instead of its folder .*
** Run: Choose File > Open > Syntax and open S05P_SPSS.sps; set CD to the extracted folder .*
** Run: Select the complete CD-through-H0 block and choose Run > Selection .*
** Run: Use Window to open Output Viewer; find H0 Import and Frame Check .*
** S05P supplied reproducible syntax. Edit only the CD folder path; keep the CSV beside this file.*
** Select and run the complete H0 block once per session before H1, H2 or H3.*

CD 'REPLACE_WITH_FOLDER_CONTAINING_CSV_AND_SPS'. ** ← change only this path*

```
BEGIN PROGRAM PYTHON3.
import csv
import math
import os
from decimal import Decimal, InvalidOperation
path = os.path.join(os.getcwd(), 'cedar_loop_pickups.csv')
expected = ['pickup_id', 'invitation_status', 'delay_flag', 'slow_flag', 'duration']
with open(path, newline='', encoding='utf-8-sig') as source:
    reader = csv.DictReader(source)
    if reader.fieldnames != expected:
        raise ValueError('H0 input error: CSV headers do not match the supplied data dictionary')
    rows = list(reader)
if len(rows) != 2400:
    raise ValueError('H0 input error: expected 2400 recorded pickup rows')
for line_number, row in enumerate(rows, start=2):
    if not row['pickup_id'] or row['invitation_status'] not in ('responded', 'nonresponse', 'not_invited'):
        raise ValueError('H0 input error: invalid identifier or invitation status on CSV line
{}'.format(line_number))
    for field in ('delay_flag', 'slow_flag', 'duration'):
        token = row[field]
        if token == '':
            continue
        try:
            value = Decimal(token)
            if not value.is_finite() or not math.isfinite(float(value)):
                raise InvalidOperation
        except (InvalidOperation, ValueError, OverflowError):
            raise ValueError('H0 input error: malformed nonblank {} on CSV line {}'.format(field, line_number))
    if field in ('delay_flag', 'slow_flag') and value not in (Decimal(0), Decimal(1)):
        raise ValueError('H0 input error: {} must be 0, 1 or blank on CSV line {}'.format(field,
line_number))
print('H0 CSV validation passed: 2400 rows; blank numeric tokens remain missing.')
END PROGRAM.
```

** Read the data file into SPSS (file type, name, and each variable's format).*

```
GET DATA
  /TYPE=TXT
  /FILE='cedar_loop_pickups.csv'
  /ENCODING='UTF8'
  /DELCASE=LINE
  /DELIMITERS=","
  /QUALIFIER='"'
  /ARRANGEMENT=DELIMITED
  /FIRSTCASE=2
  /VARIABLES=pickup_id A8 invitation_status A16 delay_flag F1.0 slow_flag F1.0 duration F24.15.
```

** Attach readable descriptions to variables (they appear in output tables).*

```
VARIABLE LABELS duration 'Recorded pickup duration (minutes)' delay_flag 'Delay over 50 minutes: 1 yes, 0 no'
slow_flag 'Alias of delay_flag: 1 yes, 0 no'.
```

** Set each variable's measurement level (nominal, ordinal, scale).*

```
VARIABLE LEVEL pickup_id (NOMINAL) invitation_status (NOMINAL) delay_flag (NOMINAL) slow_flag (NOMINAL) duration
(SCALE).
```

```
* Run the pending transformations now.
EXECUTE.
* Name the output block so you can find it in the Viewer.
TITLE 'H0 Import and Frame Check'.
* Frequency table: how many cases fall in each category.
FREQUENCIES VARIABLES=invitation_status /ORDER=ANALYSIS.
```

BEFORE THE TIMER · INSTRUCTOR DEMO

Follow along: in SPSS, run the **H1** block and leave **H1 Invitation and Recorded Delay Frequencies** open.

SPSS BLOCK FOR THIS PART · ALREADY IN YOUR SYNTAX FILE

You do not type this: it is already in `S05P_SPSS.sps`, the one syntax file you use all session. Once per SPSS session, run the `CD` line and the H0 block first (the setup code is above). Then press **Ctrl+F**, search `TITLE 'H1'`, select from the first comment line down to the last period, and choose **Run > Selection**. Green lines start with `*`: they are comments that explain the command below them, and SPSS ignores them. Every command ends with a period.

```
* Lecture connection: Nonresponse Sensitivity Bounds .
* Why: Inspect invited statuses and observed delay outcomes that underpin Part A's denominators .
* How: FREQUENCIES lists exact invitation_status categories and recorded delay_flag values; blank flags remain missing .
* How: Do not equate all missing flags with invited nonresponse: most recorded rows were not invited .
* Read the result: Read status counts separately from the valid 0/1 delay count; Excel calculates the invitation scenarios .
* Watch out: Mistaking 2,300 missing delay flags for 2,300 invited nonrespondents .
* Run: In Syntax Editor, find H1 and select its complete titled block .
* Run: Choose Run > Selection; use Window to reach Output Viewer .
* Run: Find H1 Invitation and Recorded Delay Frequencies and inspect both tables .
TITLE 'H1 Invitation and Recorded Delay Frequencies'.
* Frequency table: how many cases fall in each category.
FREQUENCIES VARIABLES=invitation_status /ORDER=ANALYSIS.
* Make the next transformation apply to the next procedure only.
TEMPORARY.
* Keep only the rows that meet the condition (others are deleted).
SELECT IF (invitation_status = 'responded' AND (delay_flag = 0 OR delay_flag = 1)).
* Frequency table: how many cases fall in each category.
FREQUENCIES VARIABLES=delay_flag /ORDER=ANALYSIS.
```

OPEN

Open `'IN-CLASS_Lab Part A'!B7`.

FILL IN THE YELLOW CELLS

#	Cell	Type this	In math	Check
1	B7	<code>=COUNTIF(Pickups[invitation_status],"responded")+COUNTIF(Pickups[invitation_status],"nonresponse")</code>	Invited pickups	125
2	B8	<code>=COUNTIFS(Pickups[invitation_status],"responded",Pickups[delay_flag],0,Pickups[delay_flag],"<>")+COUNTIFS(Pickups[invitation_status],"responded",Pickups[delay_flag],1)</code>	Eligible respondents	100

#	Cell	Type this	In math	Check
3	B9	=COUNTIFS(Pickups[invitation_status],"responded",Pickups[delay_flag],1)	Delayed respondents	24
4	B10	=B9/B8	Respondent delay rate	24.0%
5	B11	=B9/B7	Best-case invitation bound	19.2%
6	B12	=(B9+B7-B8)/B7	Worst-case invitation bound	39.2%
7	B14 B15	B14: pick the best statement from the dropdown. B15: Your reason: identify the respondent denominator, invitation denominator and what remains unobserved.		

✓ **Done when:** Cells B7:B12 display respondent proportion 24.0% and sensitivity bounds [19.2%, 39.2%]. Save.

⚠ **Watch out:** Assuming an internal transactional database automatically includes all customers without duplicate or canceled orders. Treating the 24.0% respondent rate as the true rate while ignoring the 25 missing outcomes.

Part B · Duration Estimation and Confidence Intervals

From slide 46 → work this card → Back to slide 47

· bring: The 95% CI bounds [39.62, 44.38] minutes and comparison to the 40-minute benchmark.

WHAT THE SESSION COVERED · SECTION 2 SAMPLING DISTRIBUTIONS, STANDARD ERROR, AND MEAN ESTIMATION

- **Standard Error of the Mean:** The standard error measures the expected variability of sample means across repeated samples of size n drawn from the same population. *Dividing the standard deviation by sample size n instead of the square root of n .*
- **Student t Confidence Interval for a Mean:** A Student t confidence interval provides a range of plausible values for an unknown population mean when population spread is estimated by s . *Claiming that 95 percent of individual customer pickup times fall within this narrow interval.*

COURSE SLIDES

Slides 29–46

What Standard Error Means in Business Operations 30 · Standard Error of the Mean 31–32 · the Central Limit Theorem and the Square-Root Rule 33 · Understanding Confidence Intervals for Operational... 35 · One-Sample Student t Confidence Interval for a... 36–38 · Excel 365 and SPSS Syntax for Continuous Mean... 39–41

CONCISE TOOLKIT

pp. 10–17

Standard Error of the Mean p. 11 · Cedar Loop Grocery Pickup p. 13

FULL TOOLKIT

pp. 25–43

Standard Error of the Mean p. 28 · Cedar Loop Grocery Pickup p. 27

THE RULES YOU ARE APPLYING

STANDARD ERROR OF THE MEAN

$$SE_x = \frac{s}{\sqrt{n}}$$

STUDENT T CONFIDENCE INTERVAL FOR A MEAN

$$\bar{x} \pm t^* \cdot \frac{s}{\sqrt{n}}, \quad df = n - 1$$

Just opened SPSS? Run the setup first (the `CD` line and H0; code in Part A).

BEFORE THE TIMER · INSTRUCTOR DEMO

Follow along: in SPSS, run the **H2** block and leave **H2 Mean Duration Interval** open.

SPSS BLOCK FOR THIS PART · ALREADY IN YOUR SYNTAX FILE

You do not type this: it is already in `S05P_SPSS.sps`, the one syntax file you use all session. Once per SPSS session, run the `CD` line and the H0 block first (setup code in Part A). Then press **Ctrl+F**, search `TITLE 'H2'`, select from the first comment line down to the last period, and choose **Run > Selection**. Green lines start with `*`: they are comments that explain the command below them, and SPSS ignores them. Every command ends with a period.

- * Lecture connection: Student t Confidence Interval for a Mean .
- * Why: Cross-check Part B's interval for average observed pickup duration .
- * How: EXAMINE uses nonmissing duration in minutes; DESCRIPTIVES requests valid N, mean, sample SD, SE and 95% confidence limits .
- * How: The interval estimates a mean, using sample spread rather than known population sigma .
- * Read the result: Read H2 Descriptives' mean limits in minutes and compare with B10:B11 to displayed precision .
- * Watch out: Confusing the mean interval with the range covering individual pickups .
- * Run: After H0, find H2 in Syntax Editor and select its complete titled block .
- * Run: Choose Run > Selection; switch through Window to Output Viewer .
- * Run: Read H2 Mean Duration Interval, especially Descriptives and its mean limits .

TITLE 'H2 Mean Duration Interval'.

* Explore a variable: percentiles, spread and plots, optionally by group.

```
EXAMINE VARIABLES=duration
/PLOT=NONE
/STATISTICS=DESCRIPTIVES
/CINTERVAL 95
/MISSING=LISTWISE.
```

OPEN

Open 'IN-CLASS_Lab Part B'!B7 .

FILL IN THE YELLOW CELLS

#	Cell	Type this	In math	Check
1	B7	=AVERAGE(Pickups[duration])	Observed mean duration	42.00
2	B8	=STDEV.S(Pickups[duration])	Sample duration SD	12.00
3	B9	=B8/SQRT(COUNT(Pickups[duration]))	Standard error	1.20
4	B12	=T.INV.2T(0.05,COUNT(Pickups[duration])-1)	Two-sided t multiplier	1.9842
5	B10	=B7-B12*B9	Lower 95% mean limit	39.62
6	B11	=B7+B12*B9	Upper 95% mean limit	44.38
7	B14 B15	B14: pick the best statement from the dropdown. B15: Your reason: describe what SE measures, compare the mean interval with the target and state one limitation.		

✅ **Done when:** Cells B10:B11 show lower bound 39.62 minutes and upper bound 44.38 minutes. **Save.**

⚠️ **Watch out:** Dividing the standard deviation by sample size n instead of the square root of n. Claiming that 95 percent of individual customer pickup times fall within this narrow interval.

Part C · Proportion Confidence Interval and Sample Size Planning

From slide 66 → work this card → Back to slide 70

· bring: The 438-order sample size requirement and the decision guardrail memo recommendation.

WHAT THE SESSION COVERED · SECTION 3 PROPORTION ESTIMATION, SAMPLE SIZE PLANNING, AND DECISION GUARDRAILS

- **Confidence Interval for a Population Proportion:** A proportion confidence interval bounds the true population share of binary events using the normal approximation. *Reporting the margin of error in percentage terms without clarifying percentage points.*
- **Sample Size Planning for Precision:** Sample size planning calculates the minimum number of completed units needed to guarantee a specified margin of error at a given confidence level. *Rounding down to the nearest whole integer instead of always rounding up to the next integer.*

COURSE SLIDES

Slides 48–66

Meaning of a Population Proportion and Point... 49 · One-Sample Confidence Interval for a Population... 51–53 · Constructing the Cedar Loop Confidence Interval 54 · What sample size planning means in business... 56 · Sample Size Formula for a Population Proportion 57–60 · Worked sample size calculation for Cedar Loop... 61

CONCISE TOOLKIT

pp. 18–26

Cedar Loop Grocery Pickup p. 20

FULL TOOLKIT

pp. 44–61

Confidence Interval for a Population Proportion p. 48 · Cedar Loop Grocery Pickup p. 45

THE RULES YOU ARE APPLYING

CONFIDENCE INTERVAL FOR A POPULATION PROPORTION

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

SAMPLE SIZE PLANNING FOR PRECISION

$$n = \frac{(z^*)^2 \cdot p \cdot (1-p)}{E^2}$$

Just opened SPSS? Run the setup first (the **CD** line and H0; code in Part A).

BEFORE THE TIMER · INSTRUCTOR DEMO

Follow along: in SPSS, run the **H3** block and leave **H3 Recorded Delay Frequencies** open.

SPSS BLOCK FOR THIS PART · ALREADY IN YOUR SYNTAX FILE

You do not type this: it is already in **S05P_SPSS.sps**, the one syntax file you use all session. Once per SPSS session, run the **CD** line and the H0 block first (setup code in Part A). Then press **Ctrl+F**, search **TITLE 'H3'**, select from the first comment line down to the last period, and choose **Run > Selection**. Green lines start with *****: they are comments that explain the command below them, and SPSS ignores them. Every command ends with a period.

* Lecture connection: Confidence Interval for a Population Proportion .

* Why: Cross-check the eligible observed 0/1 inputs for Part C's Excel proportion interval .

* How: FREQUENCIES displays the recorded delay_flag 0 and 1 rows and valid N; blanks remain missing .

* How: Excel, not this SPSS block, applies the normal interval and completed-sample planning formula .

* Read the result: Read valid 0/1 counts; compare their total with the observed-outcome denominator in Part C .

* Watch out: Using /FORMAT=NOTABLE, which suppresses the frequency rows needed here .

* Run: After H0, find H3 in Syntax Editor and select its complete titled block .

* Run: Choose Run > Selection; switch through Window to Output Viewer .

* Run: Read H3 Recorded Delay Frequencies, including both 0 and 1 rows .

TITLE 'H3 Recorded Delay Frequencies'.

* Make the next transformation apply to the next procedure only.

TEMPORARY.

* Keep only the rows that meet the condition (others are deleted).

SELECT IF (invitation_status = 'responded' AND (delay_flag = 0 OR delay_flag = 1)).

* Frequency table: how many cases fall in each category.

FREQUENCIES VARIABLES=delay_flag /ORDER=ANALYSIS.

OPEN

Open 'IN-CLASS_Lab Part C'!B7.

FILL IN THE YELLOW CELLS

#	Cell	Type this	In math	Check
1	B7	=COUNTIFS(Pickups[invitation_status],"responded",Pickups[delay_flag],0,Pickups[delay_flag],"<>")+COUNTIFS(Pickups[invitation_status],"responded",Pickups[delay_flag],1)	Eligible recorded outcomes	100
2	B8	=COUNTIFS(Pickups[invitation_status],"responded",Pickups[delay_flag],1)	Recorded delay outcomes	24
3	B9	=B8/B7	Observed delay share	24.0%
4	B10	=B9-1.96*SQRT(B9*(1-B9)/B7)	Lower 95% share limit	15.63%
5	B11	=B9+1.96*SQRT(B9*(1-B9)/B7)	Upper 95% share limit	32.37%
6	B12	=CEILING.MATH(1.96^2*B9*(1-B9)/0.04^2)	Required completed pickups	438
7	B13 B14	B13: pick the best statement from the dropdown. B14: Your reason: distinguish completed outcomes from invitations and identify what precision cannot repair.		

✓ Done when: Cell B12 shows required sample size n = 438 orders rounded up. Save.

 **Watch out:** Reporting the margin of error in percentage terms without clarifying percentage points. Rounding down to the nearest whole integer instead of always rounding up to the next integer.

Take-home · Post-Class Reflection and Decision Revisit

At home → work this card → Submit

WHAT THE SESSION COVERED · SECTION 3 PROPORTION ESTIMATION, SAMPLE SIZE PLANNING, AND DECISION GUARDRAILS

- **Confidence Interval for a Population Proportion:** A proportion confidence interval bounds the true population share of binary events using the normal approximation. *Reporting the margin of error in percentage terms without clarifying percentage points.*
- **Sample Size Planning for Precision:** Sample size planning calculates the minimum number of completed units needed to guarantee a specified margin of error at a given confidence level. *Rounding down to the nearest whole integer instead of always rounding up to the next integer.*

COURSE SLIDES

Slides 48-66
Meaning of a Population Proportion and Point... 49 · One-Sample Confidence Interval for a Population... 51-53 · Constructing the Cedar Loop Confidence Interval 54 · What sample size planning means in business... 56 · Sample Size Formula for a Population Proportion 57-60 · Worked sample size calculation for Cedar Loop... 61

CONCISE TOOLKIT

pp. 18-26
Cedar Loop Grocery Pickup p. 20

FULL TOOLKIT

pp. 44-61
Confidence Interval for a Population Proportion p. 48 · Cedar Loop Grocery Pickup p. 45

THE RULES YOU ARE APPLYING

CONFIDENCE INTERVAL FOR A POPULATION PROPORTION

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

SAMPLE SIZE PLANNING FOR PRECISION

$$n = \frac{(z^*)^2 \cdot p \cdot (1-p)}{E^2}$$

OPEN

Open 'IN-CLASS_Lab Part C'!B15 .

FILL IN THE YELLOW CELLS

#	Cell	Build it	In math
1	B15	B15: Your own Option A versus Option B recommendation. Option A = immediate permanent staffing; Option B = a bounded audit pilot. Cite invitation bounds, both intervals, the completed-sample target, one limitation, and a condition that would change your choice.	

✓ **Done when:** Response cell contains complete recommendation evaluating Option A versus Option B. **Save.**

⚠ **Watch out:** Reporting the margin of error in percentage terms without clarifying percentage points. Rounding down to the nearest whole integer instead of always rounding up to the next integer.

Finish

Ungraded practice

Before you close the workbook

TICK EVERY BOX 0 of 16 done

Save your work

- ☐ Workbook saved under your own name (Start Here shows it), e.g. S05P_FirstName_LastName_NetID.xlsx
- ☐ Nothing to submit: this lab is ungraded practice

Part B · IN-CLASS_Lab Part B

- ☐ Cells B10:B11 show lower bound 39.62 minutes and upper bound 44.38 minutes.
- ☐ B14 statement chosen
- ☐ B15 written: Your SE and target reason

Take-home · IN-CLASS_Lab Part C

- ☐ Response cell contains complete recommendation evaluating Option A versus Option B.
- ☐ B15 written: POST-CLASS, 15 minutes: your decision revisit

Part A · IN-CLASS_Lab Part A

- ☐ Cells B7:B12 display respondent proportion 24.0% and sensitivity bounds [19.2%, 39.2%].
- ☐ B14 statement chosen
- ☐ B15 written: Your denominator-based reason

Part C · IN-CLASS_Lab Part C

- ☐ Cell B12 shows required sample size $n = 438$ orders rounded up.
- ☐ B13 statement chosen
- ☐ B14 written: Your precision reason

Whole workbook

- ☐ Every in-class *Check your work* box reads "✓ Checked"; every take-home box reads "Entry recorded"
- ☐ No #NAME?, #VALUE!, #REF!, #NUM! or #DIV/0! anywhere
- ☐ Data sheets unchanged; formulas, not pasted values

Optional practice (not graded)

Optional Python and R companions are in your package:

- S05P_Python_Exploratory.ipynb
- S05P_R_Exploratory.ipynb